OLYMPICS ANALYTICS

Dashboard — Professional Analytical Report

Power BI solution review, analytical framework, dashboard interpretation, KPI catalogue, and recommendations

Prepared from the supplied Power BI (.pbix) dashboard

Report scope: dashboard structure, analytical intent, measures and dimensions exposed in the report, page-level interpretation, usability, and recommended next steps.

Reference context: The IOC describes the Olympic Games and Olympic Movement as a global sporting ecosystem involving athletes, National Olympic Committees, International Federations, organising committees and partners. Source: Olympics.com, “What is the International Olympic Committee (IOC) and what is its mission?”

1. Executive Summary

A management-level interpretation of the supplied Olympics Power BI dashboard.

The supplied dashboard is designed as an interactive Olympic analytics product rather than a static report. Its architecture combines headline KPIs, country-level medal rankings, athlete participation trends, medal trends over time, gender composition, sport-level analysis, and season comparisons. The design supports both high-level monitoring and focused exploration through slicers and cross-filtering.

The first page functions as the executive overview. It consolidates six headline measures — Total Athletes, Total Medals, Gold Medals, Silver Medals, Total Sports, and Total Countries — and surrounds them with trend and composition visuals. The page is configured with a Summer-season analytical context and provides Year, Season, Sex/Gender, Sport and NOC selectors.

The second page shifts the analytical lens from overall performance to competitive structure: gold-medal leaders, athlete participation by country/region, Summer-versus-Winter participation, and medals by sport. The third page provides a focused gender-participation view and event distribution by sport.

From a data-analytics perspective, the strongest feature is the breadth of dimensions exposed through a compact model: time (Year), season (Season), geography (NOC/region), gender (Sex), sport (Sport), and event-level metrics. This makes the dashboard suitable for exploratory questions such as who leads medal performance, how participation changes over time, which sports contribute the most medals, and how female participation evolves.

The principal improvement opportunity is to move from descriptive reporting toward more explicit analytical storytelling. Future versions should add period-over-period change, medal efficiency, athlete-to-medal conversion, country comparisons normalized by participation, and a dedicated methodology/data-quality page.

Overall assessment: the dashboard provides a strong foundation for an academic or portfolio-grade Power BI project and demonstrates practical use of data modelling, DAX measures, interactive filtering, and multi-page visual storytelling.

Key takeaway: The dashboard already answers “what happened?” and “who/what leads?”. The next maturity step is to answer “why?”, “how efficiently?”, and “what changed versus the previous Games?”.

2. Project Overview & Analytical Objectives

Project purpose

The objective of the solution is to transform Olympic athlete, medal, country, sport, season and event information into an interactive analytical interface. The dashboard is intended to help users compare countries, understand participation patterns, examine medal distribution, and identify changes across Olympic years.

Primary analytical objectives

- Measure the scale of Olympic participation through athlete, country, sport and event indicators.
- Compare countries by total medals and gold medals.
- Track athlete participation and medal patterns over time.
- Understand the composition of participation by gender.
- Compare Summer and Winter participation.
- Identify sports with higher medal contribution.
- Enable user-driven analysis through Year, Season, Sex/Gender, Sport and NOC filters.

Analytical questions supported

Question	Dashboard mechanism
Which countries have the strongest medal performance?	Top-10 total-medal and gold-medal rankings
How has athlete participation changed?	Year-based participation trend
How are medals distributed?	Gold/Silver/Bronze composition
How does participation differ by gender?	Male/Female composition and female trend
Which sports contribute most to medals/events?	Sport-level column/pie views
How do Summer and Winter compare?	Season comparison donut

Positioning: a descriptive and exploratory analytics solution, with a clear path toward diagnostic and performance-efficiency analytics.

3. Data Model & Technical Architecture

The PBIX report exposes two principal analytical tables in its model layout: **athlete_events** and **country_definitions**. The report visuals consistently use athlete_events for measures and core dimensions, while country_definitions supplies country/region enrichment through NOC/region fields.

Model object	Observed role	Examples used by dashboard
athlete_events	Primary analytical fact-style table	Year, Season, Sex, Sport, NOC; athlete, medal, participation and event measures
country_definitions	Geographic reference / lookup table	NOC, region
athlete_events_data_dictionary	Supporting metadata object shown in model diagram	Data dictionary / documentation
country_definitions_data_dictionary	Supporting metadata object shown in model diagram	Data dictionary / documentation

Observed dimensional structure

- **Time:** Year supports trend analysis and Games-over-Games comparison.
- **Season:** Summer/Winter enables seasonal segmentation.
- **Geography:** NOC and region enable country/region ranking.
- **Demographics:** Sex/Gender enables participation composition.
- **Competition:** Sport and Total Events enable sport-level analysis.
- **Performance:** Gold, Silver, Bronze and Total Medals provide medal outcomes.

Technical interpretation

The report metadata shows reusable measures such as Total Athletes, Total Medals, Gold Medals, Silver Medals, Total Countries, Total Sports, Athletes by year, Medals By Year, Athlete Participation, Medals By sport, Female Athletes, Male Athletes, Winter Athletes, Summer Athletes and Total Events. This indicates that the dashboard is measure-driven rather than relying only on raw column aggregation.

A professional production model should preserve a clear star-schema philosophy: keep athlete/event records at the lowest practical grain, use dedicated dimensions for date, geography, sport and season where appropriate, and centralize business logic in DAX measures. This improves maintainability, performance and consistency across visuals.

4. Dashboard Navigation & Interaction Design

The supplied report contains three analytical pages. Their progression is logical: **Executive overview → comparative performance → focused participation/event analysis**.

Page	Primary purpose	Key interactions
Page 1	Executive Olympic overview	Year, Season, Sex/Gender, Sport, NOC slicers; cross-filtering across KPIs and charts
Page 2	Country, season and sport comparison	Country/region comparisons; Summer/Winter split; sport medal comparison
Page 3	Female participation and event structure	Sport-focused analysis; female participation trend; event mix by sport

Filter strategy

- **Year:** narrows analysis to an Olympic year or selected period.
- **Season:** separates Summer and Winter Games.
- **Sex/Gender:** isolates participation composition.
- **Sport:** focuses trend and event analysis on a chosen sport.
- **NOC:** supports country-specific exploration.

Usability assessment

The use of KPI cards at the top of the overview page establishes an executive reading order. Ranking charts then answer comparative questions, while trend and composition charts provide context. The dedicated slicer column makes the page usable as a self-service analysis interface.

Recommended interaction enhancement: add a visible “Reset Filters” control, a selected-filter summary, tooltip pages for country/sport context, and dynamic titles that explicitly show the active Year/Season/NOC/Sport selection.

5. Page 1 — Executive Olympic Overview

Page 1 is the dashboard's principal executive view. It contains six KPI cards, five major analytical charts, a medal-distribution visual, and five slicers. The metadata shows that the page uses a Summer-season context as its configured default.

Component	Analytical purpose	Interpretation
Total Athletes	Scale of participation	Headline participation measure; useful for trend and country comparisons
Total Medals	Overall medal output	Aggregated medal outcome across the active filter context
Gold Medals	Top-tier success	Direct indicator of first-place medal performance
Silver Medals	Second-place success	Complements gold and bronze for medal composition
Total Sports	Programme breadth	Shows how many sports are represented in the active context
Total Countries	International reach	Measures breadth of participating NOCs/countries

Trend and ranking visuals

- **Athlete Participation Over the years:** line chart for temporal participation movement.
- **Top 10 Countries By Total Medals:** clustered bar chart for relative country performance.
- **Medals Over Years:** ribbon chart for changing medal positions/volumes across years.
- **Medal Distribution:** pie chart separating Gold, Silver and Bronze.
- **Male vs Female Athletes:** donut chart for participation composition.

Analytical value

This page supports an executive workflow: first establish the scale of the Games, then compare leading countries, inspect historical change, and finally understand the composition of participation and medals. The combination of cards, rankings, trends and composition is appropriate for a portfolio dashboard because it demonstrates multiple visual reasoning patterns rather than repeating a single chart type.

Potential advanced insight layer

- Add **Medals per 1,000 athletes** to distinguish scale from efficiency.
- Add **Gold share of total medals** to identify quality of medal performance.
- Add **YoY / Games-over-Games participation change** to quantify trend direction.
- Add **Top improving countries** using previous-Games comparisons.

6. Page 2 — Country, Season & Sport Performance

Page 2 moves beyond the executive summary and focuses on competitive structure. Four visuals are present: top countries by gold medals, athlete participation by country/region, Summer-versus-Winter athlete composition, and medals by sport.

Visual	Question answered	Professional interpretation
Top 10 Countries By Gold Medals	Who leads in gold?	Measures elite first-place performance and highlights the strongest gold-medal portfolios.
Athlete Participation by Country/Region	Where is participation concentrated?	Shows the geographic distribution of athlete participation.
Winter vs Summer Athletes	How does season shape participation?	Separates the two Olympic programme contexts.
Medals By Sport	Which sports contribute medals?	Identifies sports with larger medal volumes and supports sport portfolio analysis.

Recommended comparative metrics

- **Gold-to-total-medal ratio:** a quality indicator that rewards gold performance.
- **Medals per athlete:** participation-normalized performance.
- **Country share of total medals:** concentration analysis.
- **Sport medal share:** identifies dominant sports within the selected Games.

Interpretation caution

Raw medal totals should not be interpreted as pure athletic efficiency. Countries with larger delegations, more qualifying opportunities, or strong representation across many sports can naturally accumulate more medals. A mature analytical report should therefore pair absolute rankings with normalized measures and programme-size context.

7. Page 3 — Female Participation & Event Structure

Page 3 is a focused analytical page. Its area chart tracks Female Athletes over Year and is configured around a sport-level view; the report metadata shows a default Sport context of **Canoeing** for this visual. A second pie chart shows Total Events by Sport.

Component	Analytical objective	Suggested interpretation
Female Athletes Participation over Years	Longitudinal gender participation	Evaluate how female athlete representation changes across Olympic years.
Total Events by Sport	Programme/event structure	Understand how the Olympic programme is distributed across sports.

Why this page matters

Gender participation is a strategic dimension of Olympic analytics because participation growth can be examined independently of medal outcomes. A trend view helps distinguish one-off Games effects from sustained changes across multiple editions.

Recommended extension

- Add female participation as a percentage of total athletes rather than only absolute counts.
- Add female-to-male participation ratio and Games-over-Games percentage change.
- Allow multi-sport comparison rather than a single default sport.
- Add event-level participation where the underlying grain supports it.

Reference: Olympics.com, “Sports, programme and results” and IOC statistical publications. These should be treated as authoritative validation benchmarks when final dashboard figures are published.

8. KPI Catalogue & Analytical Logic

The following catalogue translates the measures visible in the PBIX report into professional analytical definitions. Exact DAX formulas are not exposed in the report layout metadata, so the definitions below describe intended business meaning rather than asserting the unseen implementation.

KPI / Measure	Business definition	Primary dimensions
Total Athletes	Count of athletes represented in the active analytical context.	Year, Season, Sex, Sport, NOC
Total Medals	Aggregate Olympic medal count in the active context.	Year, Season, Sport, NOC
Gold Medals	Count of gold-medal outcomes.	Year, Season, Sport, NOC
Silver Medals	Count of silver-medal outcomes.	Year, Season, Sport, NOC
Total Countries	Count of participating NOCs/countries in context.	Year, Season, Sport
Total Sports	Count of sports represented in context.	Year, Season
Athletes by year	Athlete participation aggregated by Year.	Year, Season
Medals By Year	Medal outcomes aggregated by Year.	Year, Season
Female Athletes	Female athlete participation in context.	Year, Sport, Season
Male Athletes	Male athlete participation in context.	Year, Sport, Season
Winter Athletes	Participation under Winter season context.	Season, Year
Summer Athletes	Participation under Summer season context.	Season, Year
Total Events	Event count in the active sport context.	Sport, Year, Season

Suggested advanced DAX layer

For a professional portfolio release, add measures such as **Medal Efficiency = Total Medals / Total Athletes**, **Gold Rate = Gold Medals / Total Medals**, **Female Participation % = Female Athletes / Total Athletes**, **Medal Share % = Country Medals / All Medals**, and period-over-period change measures. These measures should use DIVIDE() for safe handling of zero denominators and should respect slicer context.

9. Analytical Insights, Business Value & Storytelling

Because the supplied PBIX contains the dashboard definition but not an externally exported result table, this report deliberately avoids inventing numeric findings. Instead, it identifies the analytical insights the dashboard is structured to surface.

Insight category	What the dashboard can reveal	Decision value
Competitive leadership	Top countries by total medals and gold medals.	Benchmarking national performance.
Participation growth	Athlete participation trend by year.	Understanding expansion/contraction of participation.
Medal composition	Gold/Silver/Bronze mix.	Distinguishing overall volume from top-tier success.
Geographic concentration	Athlete participation by country/region.	Identifying participation concentration and breadth.
Sport contribution	Medals and events by sport.	Understanding which sports drive medal volume.
Gender participation	Male/female composition and female trend.	Assessing participation balance over time.
Seasonal structure	Summer versus Winter participation.	Separating distinct Olympic programme contexts.

Recommended storytelling sequence

- Start with **scale**: athletes, countries, sports and medals.
- Move to **leaders**: total medals and gold medals by country.
- Explain **change**: participation and medals across years.
- Explain **composition**: medal types, gender and seasons.
- Drill into **drivers**: sports and events.
- Finish with **efficiency**: medals relative to participation.

This sequence turns a collection of charts into an analytical narrative and is especially effective for interviews, academic evaluation, and portfolio presentations.

10. Data Quality, Governance & Limitations

A professional dashboard should distinguish between what is visually available and what has been formally validated. The supplied PBIX was inspected at the report-definition level: page structure, visual types, fields, measures, filters and model-layout objects were reviewed. The embedded Power BI DataModel is stored in a compressed internal format and was not independently recalculated in this environment.

Area	Current assessment	Recommended control
Source authority	Dashboard should be reconciled to authoritative Olympic results.	Validate against IOC/Olympics results and published statistical handbooks.
Medal changes	Historical medal tables can change after disqualifications/reallocations.	Document refresh date and medal-reallocation treatment.
Country/NOC mapping	Country definitions are handled through a reference table.	Maintain a versioned NOC/region mapping and document historical naming changes.
Duplicate athletes	Athlete-level datasets may contain repeated appearances across events/Games.	Define whether KPIs mean rows, athlete-Games, unique athletes, or athlete-event records.
Season comparisons	Summer and Winter are analytically different programmes.	Avoid direct efficiency comparisons without programme-size context.
Default filters	Some visuals have configured defaults such as Summer and a sport context.	Expose active filters clearly to reduce interpretation risk.

Important metric-definition issue

“Total Athletes” is especially important to define. A row-level count, distinct athlete count, athlete-Games count, or athlete-event count can produce materially different results. For a professional dashboard, the KPI tooltip or methodology page should explicitly state the counting rule.

Reference note: IOC statistical publications and official-results documentation show that Olympic results require careful historical treatment, including cases where official results and later medal allocations can differ. Validate the dashboard against authoritative Olympic records before publishing headline numbers.

11. Professional Recommendations & Roadmap

Priority 1 — Strengthen analytical depth

- Add medal efficiency, gold rate, medal share, and participation-normalized metrics.
- Add Games-over-Games and rolling trend calculations.
- Add country performance decomposition by sport.

Priority 2 — Improve executive usability

- Create a dynamic subtitle showing active Year / Season / NOC / Sport filters.
- Add Reset Filters and a Selected Filters card.
- Use report-page tooltips for country and sport drill-downs.

Priority 3 — Improve data governance

- Publish a data dictionary for every KPI.
- Document the source dataset, refresh date, NOC mapping and medal reallocation rules.
- Validate headline numbers against official Olympic sources before publication.

Priority 4 — Improve portfolio presentation

- Add a dedicated “Methodology” page describing model grain and DAX logic.
- Add a “Key Insights” page with three to five quantified findings.
- Include accessibility checks: contrast, alt text, readable labels and keyboard-friendly navigation.

Phase	Deliverable	Outcome
1	Metric governance + validation	Trusted KPI definitions
2	Advanced DAX measures	Efficiency and trend analytics
3	UX enhancements	Faster self-service exploration
4	Insight narrative	Executive-ready storytelling
5	Refresh & QA process	Maintainable production dashboard

12. Conclusion

The supplied Olympics Analytics Dashboard is a well-structured Power BI solution with a clear three-page analytical journey. It combines KPI monitoring, country ranking, participation trends, medal composition, gender analysis, seasonal comparison, and sport-level analysis in a single interactive environment.

Its strongest qualities are breadth of analytical dimensions, effective use of multiple visual types, reusable measure-driven reporting, and a logical progression from overview to focused analysis. These characteristics make it suitable as a strong academic project and a credible data-analytics portfolio piece.

To reach a professional production standard, the dashboard should evolve from descriptive reporting toward normalized performance measures, explicit period comparisons, stronger methodology documentation, authoritative validation, and more visible filter state. The recommended roadmap preserves the existing dashboard structure while adding analytical rigor rather than unnecessary visual complexity.

Final assessment

Dimension	Assessment	Comment
Analytical coverage	Strong	Covers participation, medals, geography, gender, season and sport.
Dashboard structure	Strong	Three-page progression supports executive-to-detail analysis.
Interactivity	Strong	Multiple slicers and cross-filtering enable self-service analysis.
Metric governance	Moderate	Definitions should be documented explicitly.
Advanced analytics	Moderate	Normalization and period-over-period measures are the main next step.
Production readiness	Good foundation	Requires validation, QA and refresh lineage for publication.

Overall conclusion: A strong foundation with clear potential to become an interview-ready, executive-quality Olympic performance analytics product.

Appendix A — Visual Inventory Extracted from the PBIX

The inventory below is based on the report-definition metadata contained in the supplied .pbix file. It is intended as a technical appendix for project documentation.

Page 1

Type	Visual / element
KPI	Total Athletes
KPI	Total Medals
KPI	Gold Medals
KPI	Silver Medals
KPI	Total Sports
KPI	Total Countries
Line chart	Athlete Participation Over the years
Donut chart	Male vs Female Athletes
Clustered bar	Top 10 Countries By Total Medals
Ribbon chart	Medals Over Years
Pie chart	Medal Distribution
Slicers	Year, Season, Sex/Gender, Sport, NOC

Page 2

Type	Visual / element
Clustered column	Top 10 Countries By Gold Medals
Clustered bar	Athlete Participation by Country/Region
Donut chart	Winter vs Summer Athletes
Clustered column	Medals By Sport

Page 3

Type	Visual / element
Area chart	Female Athletes Participation over Years
Pie chart	Total Events by Sport
Page filter/default	Sport = Canoeing in the female-participation view

Appendix B — External Reference Context

Reference sources consulted for this report: Olympics.com / IOC Support — “Sports, programme and results”; Olympics.com / IOC Support — “What is the International Olympic Committee (IOC) and what is its mission?”; Olympic Studies Centre — statistical and historical Olympic reference publications; IOC historical-results documentation. These sources provide context and validation benchmarks rather than replacing the data contained in the supplied dashboard.

Source note: This report is a professional interpretation of the supplied Power BI report definition. Numeric findings should be populated directly from the live PBIX or an exported result table rather than inferred from chart design alone.